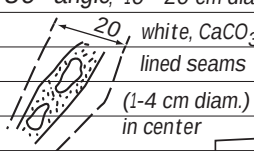


**DRAFT 3/2002**

|                                                                                                                                |  |                                       |  |                                                                                                                |  |                                                                                                         |  |                                                                                                                                                 |  |                                                                                                                  |  |                                                                              |  |                                                                 |  |                                                                               |  |
|--------------------------------------------------------------------------------------------------------------------------------|--|---------------------------------------|--|----------------------------------------------------------------------------------------------------------------|--|---------------------------------------------------------------------------------------------------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------|--|-----------------------------------------------------------------|--|-------------------------------------------------------------------------------|--|
| Series or Component Name:<br><i>Caveat Emptor</i>                                                                              |  |                                       |  | Map Unit Symbol:<br><i>CaC</i>                                                                                 |  | Photo #:<br><i>127A</i>                                                                                 |  | Classification:<br><i>fine, smectitic, mesic, Typic Argiudoll</i>                                                                               |  |                                                                                                                  |  | Soil Moist. Regime (Tax.):<br><i>Udic (≈ 27" annual)</i>                     |  |                                                                 |  |                                                                               |  |
| Describer(s):<br><i>PJS &amp; DAW</i>                                                                                          |  | Date:<br><i>10/12/2000</i>            |  | Weather:<br><i>sunny</i>                                                                                       |  | Temp.: Air: <i>78 °F</i><br>Soil: _____ Depth: _____                                                    |  | Latitude: <i>40 ° 49 ' 10.0 " N</i><br>Longitude: <i>96 ° 46 ' 06.1 " W</i>                                                                     |  | Datum:<br><i>NAD '83</i>                                                                                         |  | Location: <i>NE 1/4, SW 1/4</i><br>Sec. <i>20</i> T. <i>10N</i> R. <i>6E</i> |  |                                                                 |  |                                                                               |  |
| UTM: Zone : _____ mE: _____ mN: _____                                                                                          |  |                                       |  | Topo Quad: <i>Emerald, NE</i><br><i>7.5 topo 1978</i>                                                          |  | Site ID: Yr: State: _____<br><i>S 2002 NE - 109 - 006</i>                                               |  | County: Pedon #: _____<br><i>Lancaster, Co</i>                                                                                                  |  | Soil Survey Area:<br><i>106</i>                                                                                  |  | MLRA / LRU:<br><i>106</i>                                                    |  | Transect: ID: <i>2</i><br>Stop #: <i>6</i> Interval: <i>10m</i> |  |                                                                               |  |
| Landscape:<br><i>upland</i>                                                                                                    |  | Landform:<br><i>low hill</i>          |  | Microfeature:<br>_____                                                                                         |  | Anthro:<br><i>furrow</i>                                                                                |  | Elevation:<br><i>1240 '</i>                                                                                                                     |  | Aspect:<br><i>34 °E</i>                                                                                          |  | Slope (%):<br><i>6%</i>                                                      |  | Slope Complexity:<br><i>simple</i>                              |  | Slope Shape: ( <i>Up &amp; Dn / Across</i> )<br><i>V V ( convex, convex )</i> |  |
| Hillslope Profile Position:<br><i>shoulder</i>                                                                                 |  | Geom. Component:<br><i>nose slope</i> |  | Microrelief:<br><i>micro-low</i>                                                                               |  | Physio. Division:<br><i>Interior Plains</i>                                                             |  | Physio. Province:<br><i>Central Lowland</i>                                                                                                     |  | Physio. Section:<br><i>Dissected Till Plain</i>                                                                  |  | State Physio. Area:<br>_____                                                 |  | Local Physio. Area:<br><i>Middle Creek Basin</i>                |  |                                                                               |  |
| Drainage: <i>WD</i><br><i>(Well Drained)</i>                                                                                   |  | Flooding:<br><i>none</i>              |  | Ponding:<br><i>none</i>                                                                                        |  | Soil Moisture Status:<br><i>moist</i>                                                                   |  | Permeability: <i>slow (est.)</i><br><i>low (ave.= 0.011 cm/hr;</i><br><i>K<sub>sat</sub>: n=3; depth: 35–50 cm)</i>                             |  |                                                                                                                  |  | Land Cover / Use:<br><i>Soybeans (CRC)</i>                                   |  |                                                                 |  |                                                                               |  |
| Parent Material: <i>loess, over</i><br><i>pedisediment, over till</i>                                                          |  |                                       |  | Bedrock: Kind: _____ Fract.: _____ Hard.: _____ Depth: _____<br><i>Sandstone (in adjacent gully) &gt; 15 m</i> |  |                                                                                                         |  | Lithostrat. Units: _____ Group: _____ Formation: _____ Member: _____<br><i>Peoria Loess, Gillman Canyon, unnamed pedisediment, unnamed till</i> |  |                                                                                                                  |  |                                                                              |  |                                                                 |  |                                                                               |  |
| Erosion: <i>R</i><br><i>(Rill)</i>                                                                                             |  | Kind: _____ Degree: <i>1</i>          |  | Runoff: <i>VH</i><br><i>(Very High)</i>                                                                        |  | Surface Frag %: _____ GR: _____ CB: _____ ST: _____ BD: _____ CN: _____ FL: _____<br>Kind: <i>– 0 –</i> |  |                                                                                                                                                 |  | Diagnostic Horz. / Prop.: _____ Kind: <i>mollic</i> Depth: <i>0 – 30 cm</i><br><i>argillic</i> <i>30 – 90 cm</i> |  |                                                                              |  |                                                                 |  |                                                                               |  |
| P. S. Control Section : Ave. Clay %: _____ Ave. Rock Frag %: _____<br>Depth Range: <i>30 – 80 cm</i> <i>37%</i> <i>&lt; 1%</i> |  |                                       |  |                                                                                                                |  |                                                                                                         |  |                                                                                                                                                 |  |                                                                                                                  |  |                                                                              |  |                                                                 |  |                                                                               |  |

[illegible]

| MISCELLANEOUS FIELD NOTES / SKETCH :                                               |                    |                                    |        |     |
|------------------------------------------------------------------------------------|--------------------|------------------------------------|--------|-----|
| deep polygonal joints in till:                                                     |                    | M                                  | cm     |     |
| ≈ 30° angle, 10 - 20 cm diam.                                                      |                    | 0                                  | 0      |     |
|  | Peoria Loess       | loess                              | Ap     | 0   |
|                                                                                    |                    | white, CaCO <sub>3</sub>           | A      | 20  |
|                                                                                    |                    | lined seams                        | Bt 1   | 30  |
|                                                                                    |                    | (1-4 cm diam.) in center           | Bt 2   | 60  |
| bounded by brown / black                                                           | Gillman Canyon     | colluvial loess with scattered gr. | 1      | 90  |
| Fe / Mn zone which is                                                              | pedisediment       | colluvial loess                    | 2 Bt 3 | 130 |
| bounded by                                                                         |                    | red paleosol                       | 3B     | 145 |
| yellow / br.                                                                       |                    | weathered,                         | 4Bt 1  | 160 |
| Fe (goethite)                                                                      |                    | oxidized                           | 4Bt 2  | 210 |
| halo at                                                                            | Pre-Illinoian till | mottled, unoxidized,               | 2      | 230 |
| 250 cm                                                                             |                    | leached                            | 4 BC   | 260 |
| depth.                                                                             |                    | mottled, unoxidized,               | 4C     |     |
|                                                                                    |                    | unleached                          |        |     |

| Component Name: |                    |            |      |               | Map Unit Symbol: |                                |                 |                                               |    |           |                               |      |             |               | Date:   |            |             |                                                                  |    |     |     |    |     |  |
|-----------------|--------------------|------------|------|---------------|------------------|--------------------------------|-----------------|-----------------------------------------------|----|-----------|-------------------------------|------|-------------|---------------|---------|------------|-------------|------------------------------------------------------------------|----|-----|-----|----|-----|--|
| Obsr. Method    | Depth<br>(in) (cm) | Horizon    | Bnd  | Matrix Color  |                  | Texture                        | Rock Frags      |                                               |    | Structure |                               |      | Consistence |               |         |            | Mottles     |                                                                  |    |     |     |    |     |  |
|                 |                    |            |      | Dry           | Moist            |                                | Kind %          | Rnd                                           | Sz | Grade     | Sz                            | Type | Dry         | Mst           | Stk     | Pls        | %           | Sz                                                               | Cn | Col | Mst | Sp | Loc |  |
| 1               | LP                 | 0 – 20     | Ap   | Abrupt Smooth | 10YR 4/2         | 10YR 3/1                       | silt loam (sil) | — 0 —                                         |    |           | common, fine & med. granular  |      |             | Slightly Hard | Friable | non-sticky | non-plastic | None                                                             |    |     |     |    |     |  |
| 2               | LP                 | 20 – 30    | A    | CW            | 10YR 4/2         | 10YR 3/1                       | sil             | — 0 —                                         |    |           | 3 f,m abk.                    |      |             | MH            | FI      | SO         | PO          |                                                                  |    |     |     |    |     |  |
| 3               | LP                 | 30 – 60    | Bt1  | GW            | 2.5 Y 6/2        | 10YR 5/3                       | sicl            | — 0 —                                         |    |           | 2 m,c sbk                     |      |             | H             | VFI     | SS         | MP          |                                                                  |    |     |     |    |     |  |
| 4               | LP                 | 60 – 90    | 2Bt2 | GW            | 10YR 6/3         | 70% 10YR 4/3<br>30% 10YR 5/3   | sicl            | 2% scattered f,m rounded gr.                  |    |           | 2 m pr ⇒<br>2 m sbk           |      |             | H             | VFI     | SS         | MP          |                                                                  |    |     |     |    |     |  |
| 5               | LP                 | 90 – 130   | 2Bt3 | AW            | 10YR 4/4         | 40% 7.5YR 4/3<br>60% 7.5YR 3/3 | sil             | 2% scattered f,m rounded gr.                  |    |           | 1 m, co. pr ⇒<br>2 m sbk      |      |             | MH            | FI      | SS         | SP          |                                                                  |    |     |     |    |     |  |
| 6               | LP                 | 130 – 145  | 3B   | AW            | 7.5YR 5/4        | 7.5YR 4/6                      | xgrscl          | 85% f, m, co. rounded gravels mixed lithology |    |           | 0 sg                          |      |             | L             | L       | SO         | SO          |                                                                  |    |     |     |    |     |  |
| 7               | LP                 | 145 – 160  | 4Bt1 | GW            | 7.5YR 5/6        | 7.5YR 4/6                      | cl              | 10% f, rounded gr., mixed lithology           |    |           | 2 m sbk ⇒<br>3 vf,f sbk       |      |             | VH            | EF      | MS         | MP          |                                                                  | ↓  |     |     |    |     |  |
| 8               | LP                 | 160 – 210  | 4BT2 | DW            | 7.5YR 4/4        | 7.5YR 4/4                      | cl              | 1% f,m rounded gr., mixed lithology           |    |           | 3 co., vco pr ⇒<br>3 f, m sbk |      |             | EH            | SR      | MS         | MP          | 15% coarse, faint, 10 YR 4/3 mottles, M, irregular, on ped faces |    |     |     |    |     |  |
| 9               | LP                 | 210 – 230  | 4BC  | DI            | 2.5YR 7/2        | 2.5YR 5/2                      | c               | 1% f,m rounded gr., mixed lithology           |    |           | 3 co, vco pr ⇒<br>2 m sbk     |      |             | EH            | R       | VS         | VP          | None                                                             |    |     |     |    |     |  |
| 10              | SP                 | 230 – 260+ | 4C   | —             | 2.5YR 7/2        | 2.5YR 5/2                      | c               | 1% f,m rounded gr., mixed lithology           |    |           | 3 co, vco pr ⇒<br>3 f,m abk   |      |             | R             | R       | VS         | VP          | None                                                             |    |     |     |    |     |  |

|    | Redoximorphic Features                                     |    |    |    |    |    |     |    | Concentrations                                                          |   |    |    |    |    |    |     | Ped / V. Surface Features                              |     |                                                            |                                                     |      |    | Roots              |                                |                              | Pores      |                   |         | pH                  | Effer  | Clay   | CCE                                                                              | Notes                                                                        |   |
|----|------------------------------------------------------------|----|----|----|----|----|-----|----|-------------------------------------------------------------------------|---|----|----|----|----|----|-----|--------------------------------------------------------|-----|------------------------------------------------------------|-----------------------------------------------------|------|----|--------------------|--------------------------------|------------------------------|------------|-------------------|---------|---------------------|--------|--------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------|---|
|    | %                                                          | Sz | Cn | Hd | Sp | Kd | Loc | Bd | Col                                                                     | % | Sz | Cn | Hd | Sp | Kd | Loc | Bd                                                     | Col | %                                                          | Dst                                                 | Cont | Kd | Loc                | Col                            | Qty                          | Sz         | Loc               | Qty     | Sz                  | Shp    | (meth) | (agent)                                                                          |                                                                              | % |
| 1  | None                                                       |    |    |    |    |    |     |    | None                                                                    |   |    |    |    |    |    |     | None                                                   |     |                                                            |                                                     |      |    | 1 m T<br>2 vf, f T | few, very fine, dendr. tubular |                              |            | 5.0               | *       | NE, H2              |        |        | (* pH by pocket pH meter, 1:1 soil to water)                                     |                                                                              |   |
| 2  |                                                            |    |    |    |    |    |     |    |                                                                         |   |    |    |    |    |    |     |                                                        |     |                                                            |                                                     |      |    |                    |                                | 1 m T<br>2 vf, f T           | 2 vf TE    |                   |         | 6.0                 |        | NE, H2 |                                                                                  |                                                                              |   |
| 3  |                                                            |    |    |    |    |    |     |    |                                                                         |   |    |    |    |    |    |     |                                                        |     |                                                            |                                                     |      |    |                    |                                | 2 vf T<br>1 f T              | 2 vf, f TE |                   |         | 6.7                 |        | NE, H2 |                                                                                  |                                                                              |   |
| 4  |                                                            |    |    |    |    |    |     |    |                                                                         |   |    |    |    |    |    |     |                                                        |     |                                                            |                                                     |      |    |                    |                                | 1 vf T<br>1 f T              | 2 vf, f TE |                   |         | 6.9                 |        | NE, H2 |                                                                                  |                                                                              |   |
| 5  |                                                            |    |    |    |    |    |     |    |                                                                         |   |    |    |    |    |    |     |                                                        |     |                                                            |                                                     |      |    |                    |                                | few, very fine, between peds |            |                   | 2 vf TE | 7.2                 |        | NE, H2 |                                                                                  |                                                                              |   |
| 6  |                                                            |    |    |    |    |    |     |    |                                                                         |   |    |    |    |    |    |     |                                                        |     | 20%, prominent, discontinuous clay films on rock fragments |                                                     |      |    |                    |                                | None                         | 3 vf, f IR | 7.1               |         | NE, H2              |        |        | gravel pavement with scattered, small gully fills                                |                                                                              |   |
| 7  |                                                            |    |    |    |    |    |     |    |                                                                         |   |    |    |    |    |    |     |                                                        |     | 85%, P, cont. (C), clay films (CLF) on all ped faces (PF)  |                                                     |      |    |                    |                                |                              | 2 vf, f TE | 7.1               |         | NE, H2              |        |        | till joint ghosts remain; truncated paleosol; strong argillans & pedo. structure |                                                                              |   |
| 8  | common, med., distinct 10 YR 6/3 iron depletions in matrix |    |    |    |    |    |     |    |                                                                         |   |    |    |    |    |    |     |                                                        |     |                                                            | 40%, D, discont. (D), CLF on PF                     |      |    |                    |                                |                              |            | 2 vf TE<br>1 f TE | 7.6     |                     | NE, H2 |        |                                                                                  | till joints ghost and fade upwards to top at 45°; clay & Fe/Mn coated prisms |   |
| 9  | f, 3, P, 10 YR 2/1, MNF, APE                               |    |    |    |    |    |     |    |                                                                         |   |    |    |    |    |    |     |                                                        |     |                                                            | 27%, D, patchy (P), CLF on ped and void faces (PVF) |      |    |                    |                                |                              |            | 2 vf TE           | 7.7     |                     | SL, H2 |        |                                                                                  | polygonal till joints ghost & tip 30° to North (downslope)                   |   |
| 10 | c, 4, P, 2.5 / N, MNF, on prism faces (APF)                |    |    |    |    |    |     |    | c, vco, P, white, I, CaCO3 nodules along joints (CRK) & in matrix (MAT) |   |    |    |    |    |    |     | 7%, P, discont., pressure faces (PRF) on pf throughout |     |                                                            |                                                     |      |    |                    |                                |                              | 1 vf, f IG | 8.2               |         | SL, H2 (nodules VE) |        |        | till joints tip 30° to North (down slope)                                        |                                                                              |   |